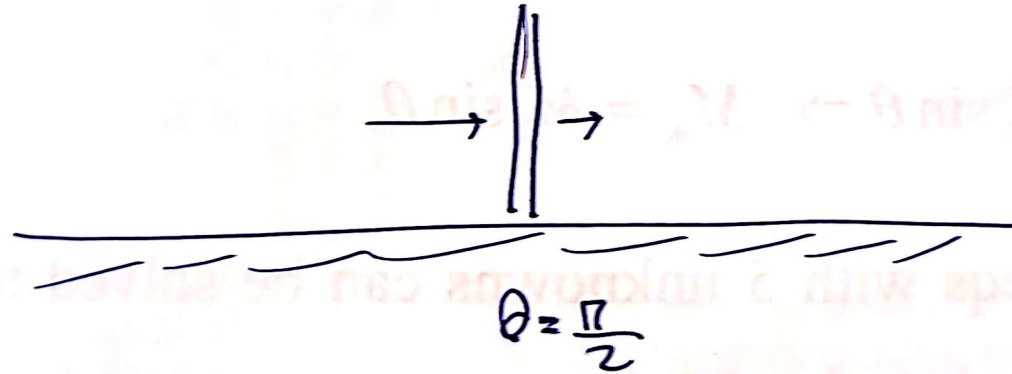


if $\delta =$



Normal Shock



M- δ - θ relation (need 2 out of 3)

known M_1 

$$\cot \delta = \tan \theta \left[\frac{(\gamma + 1) M_1^2}{2(M_{n_1}^2 - 1)} - 1 \right] \quad (8.73)$$

$M_{n_1} = M_1 \sin \theta$ we get θ

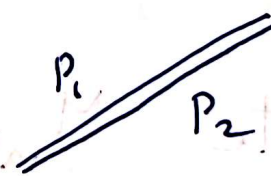
- normal shocks are a special case of oblique shocks for

$$\theta = \frac{\pi}{2}, \quad \delta = 0$$

(no deflection)

$M_1 > 1 \quad | \quad M_2 < 1$
shock

$$\frac{p_2}{p_1} = \frac{2\gamma M_{n_1}^2 - (\gamma - 1)}{\gamma + 1} = \boxed{1 + \frac{2\gamma}{\gamma + 1} (M_{n_1}^2 - 1)} \quad (8.74)$$



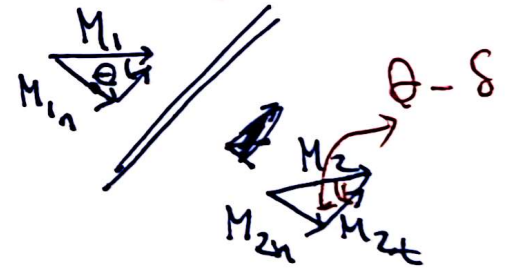
Every parameter depends on M_{n_1}

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1) M_{n_1}^2}{(\gamma - 1) M_{n_1}^2 + 2} \quad (8.75)$$

You have M_1 & $\delta \rightarrow$ get θ
get $M_{n_1} = M_1 \sin \theta \rightarrow$ get other parameters

At the end of your book, there is normal shock table, we enter that table with M_{1n} & read from it M_{2n} , then convert M_{2n} to M_2 from:

$$M_2 = \frac{M_{2n}}{\sin(\theta - \delta)}$$



Note $M_{1n} = M_1 \sin \theta$

$$M_{2n} = M_2 \sin(\theta - \delta)$$

So the first step is to read θ



$$\frac{T_2}{T_1} = \frac{p_2}{p_1} \frac{\rho_1}{\rho_2} \Rightarrow \frac{T_2}{T_1} = \frac{[2\gamma M_{n_1}^2 - (\gamma - 1)][(\gamma - 1)M_{n_1}^2 + 2]}{(\gamma + 1)^2 M_{n_1}^2} \quad (8.76)$$

$$M_{n_2}^2 = \frac{M_{n_1}^2 + [2/(\gamma - 1)]}{[2\gamma/(\gamma - 1)]M_{n_1}^2 - 1} = \frac{1 + [(\gamma - 1)/2]M_{n_1}^2}{\gamma M_{n_1}^2 - (\gamma - 1)/2}$$

$\lim_{M_{n_1} \rightarrow \infty}$

Shock table

M_{n_1}	M_{n_2}	P_2/P_1	T_2/T_1
1	1	1	1
2	0.577	0.721	1.688
3	0.475	0.475	2.669
4	0.435	0.357	3.857
5	0.415	0.298	5.041
6	0.403	0.255	6.200
7	0.397	0.225	7.325
8	0.393	0.203	8.413
9	0.390	0.187	9.464
10	0.389	0.175	10.494

The Mach number behind the shock can be found from

$$M_2 = \frac{M_{n_2}}{\sin(\theta - \delta)} \Rightarrow M_2^2 = \frac{1 + [(\gamma - 1)/2]M_{n_1}^2}{[\gamma M_{n_1}^2 - (\gamma - 1)/2] \sin^2(\theta - \delta)}$$

$M_{n_2}^2 \rightarrow \left[\frac{\gamma - 1}{2\gamma} \right]$

$M_{n_2} = \sqrt{\frac{\gamma - 1}{2\gamma}}$

$$\frac{p_{t_2}}{p_{t_1}} = e^{-(s_2 - s_1)/R} = \left[\frac{(\gamma + 1)M_{n_1}^2}{(\gamma - 1)M_{n_1}^2 + 2} \right]^{\gamma/(\gamma - 1)} \left[\frac{(\gamma + 1)}{2\gamma M_{n_1}^2 - (\gamma - 1)} \right]^{1/(\gamma - 1)} \quad (8.78)$$

$$p_{t_2} < p_{t_1}$$

Shock is impossible
in subsonic flows

$$\text{if } M_{n_1} = 1 \rightarrow \Delta s = 0 \rightarrow p_{t_2} = p_{t_1}$$

$$\text{if } M_{n_1} > 1 \rightarrow \Delta s > 0 \quad p_{t_2} < p_{t_1}$$

$$\Leftrightarrow \text{if } M_{n_1} < 1 \rightarrow \Delta s < 0 \quad p_{t_2} > p_{t_1} \text{ impossible}$$

Please note the change in entropy

$$s_2 - s_1 < 0 \quad \text{for} \quad M_{n_1} < 1$$

entropy cannot reduce
(not physical)

$$s_2 - s_1 = 0 \quad \text{for} \quad M_{n_1} = 1 \quad (\text{sound/Mach wave})$$

$$s_2 - s_1 > 0 \quad \text{for} \quad M_{n_1} > 1 \quad (\text{shock wave})$$

$$C_p = \frac{p_2 - p_1}{q} = \frac{4(M_{n_1}^2 - 1)}{(\gamma + 1)M_{n_1}^2}$$

Trends and graphs are below

if we write entropy relation

it is

$$\begin{aligned} s_2 - s_1 &= c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \\ &= c_p \ln \frac{T_{02}}{T_{01}} - R \ln \frac{p_{02}}{p_{01}} \end{aligned}$$

①
②

Note that T_0 is the same across shock
as it is adiabatic $\rightarrow$ so $s_2 - s_1 = -R \ln \frac{p_{02}}{p_{01}}$
 $p_{02} < p_{01}$ for shock waves

N.S

N.S is ^{an} 0.5 with
 $\theta = \frac{\pi}{2}$

0.5

8

TABLE 8.4 Flow Property Changes across Weak Shock Waves

Normal shock	Oblique shock
$p_2 > p_1$ $\rho_2 > \rho_1$ $T_2 > T_1$ $M_2 < 1$ $p_{t,2} < p_{t,1}$ $T_{t,2} = T_{t,1}$	$p_2 > p_1$ $\rho_2 > \rho_1$ $T_2 > T_1$ $M_2 < M_1$ $p_{t,2} < p_{t,1}$ $T_{t,2} = T_{t,1}$
	total pressure decreases (adiabatic)

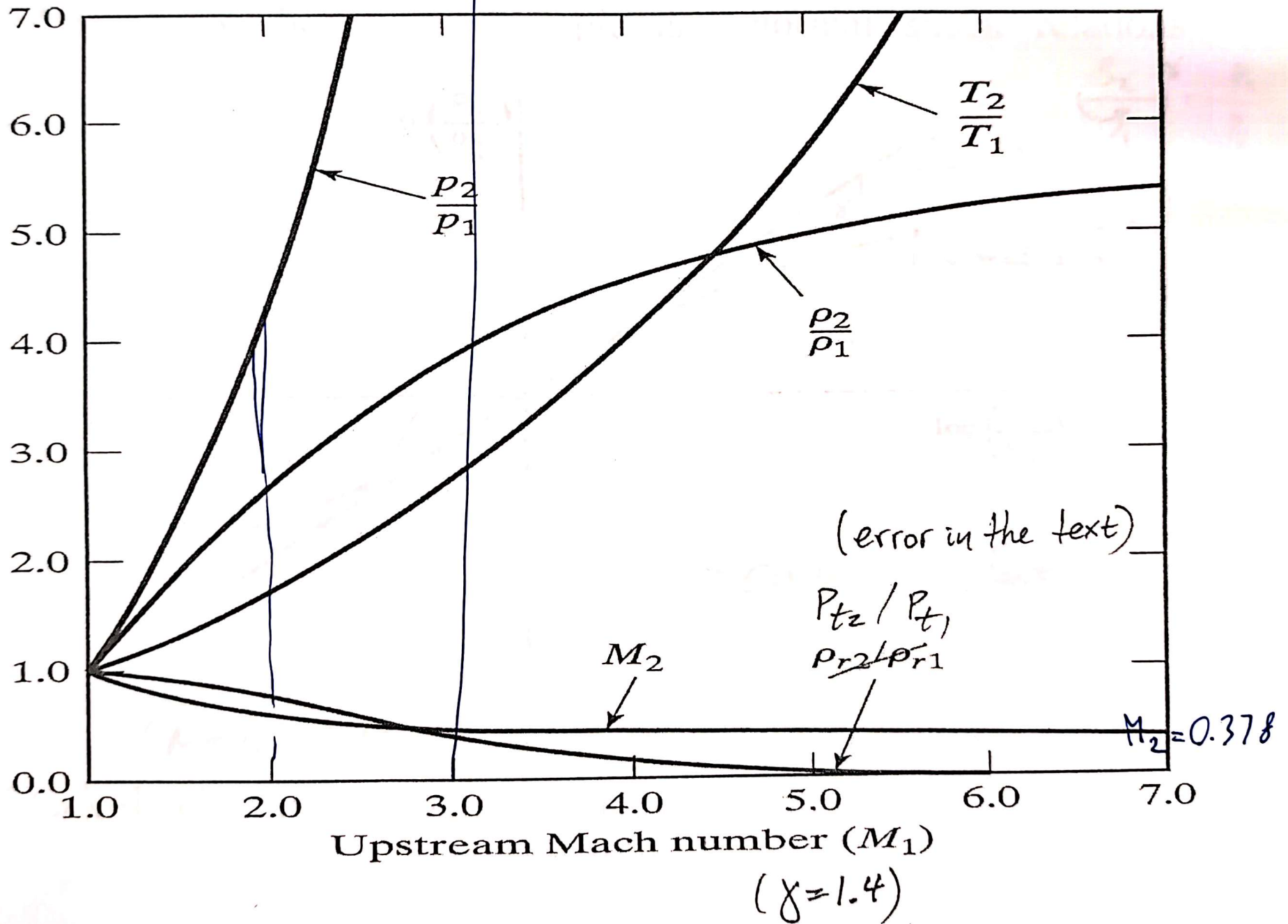
limits

for $M_1 \rightarrow \infty$ $M_2 \rightarrow \sqrt{\frac{\gamma-1}{2\gamma}} = 0.378$

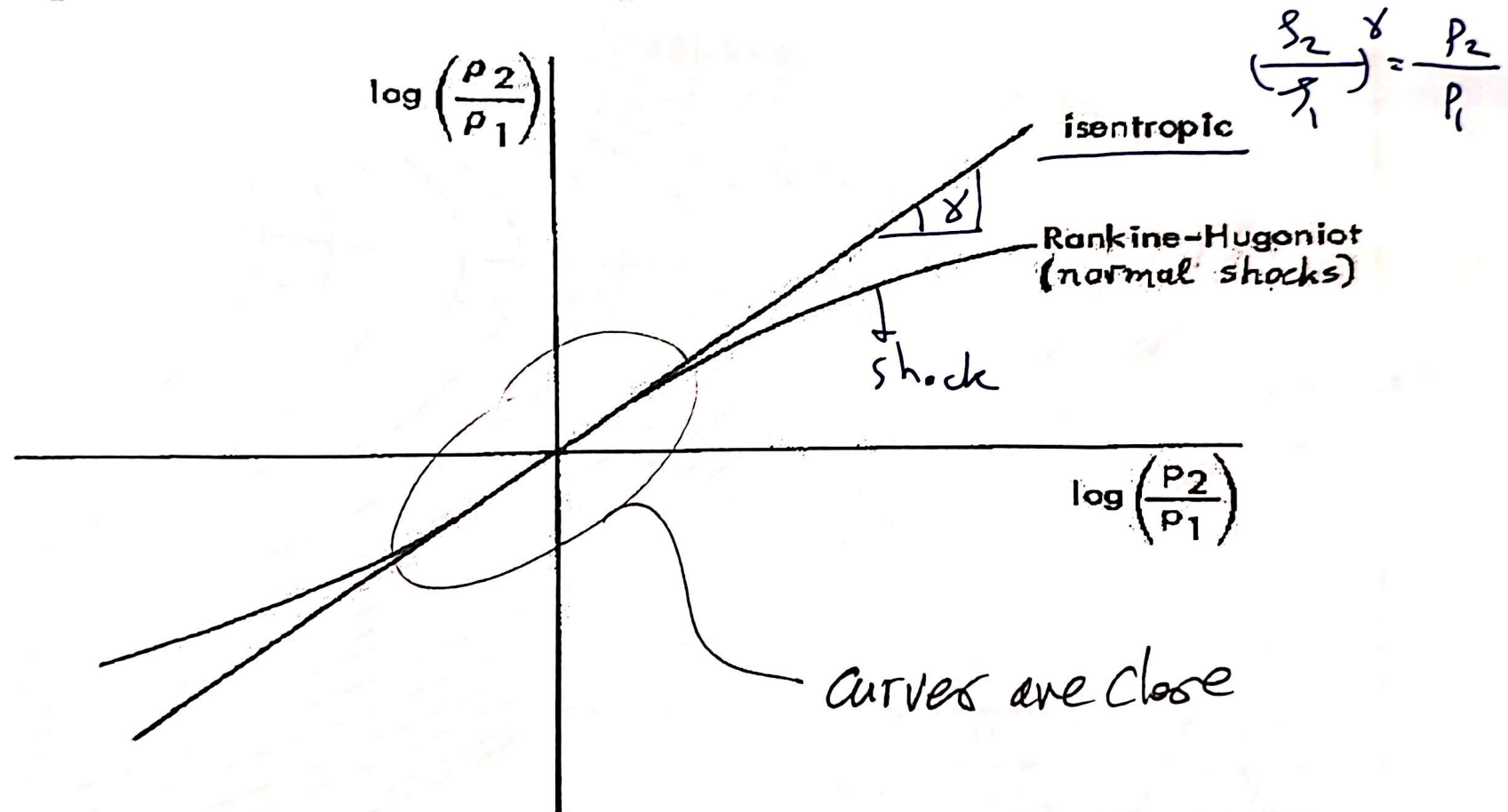
$$\frac{\rho_2}{\rho_1} \Rightarrow \frac{\gamma+1}{\gamma-1} = 6$$

(strength) $\frac{p_2}{p_1} \rightarrow \infty$

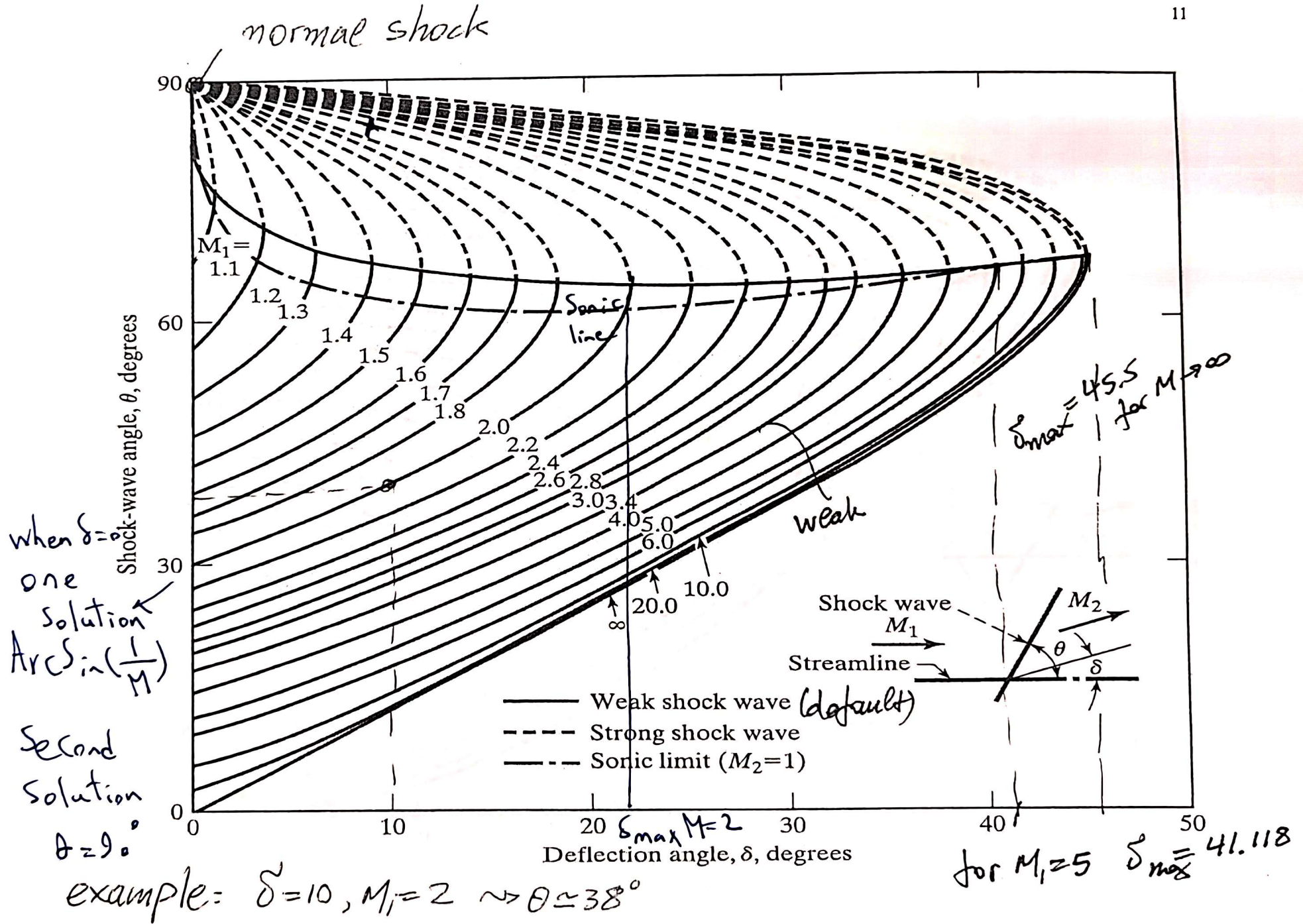
$$\frac{T_2}{T_1} \rightarrow \infty$$

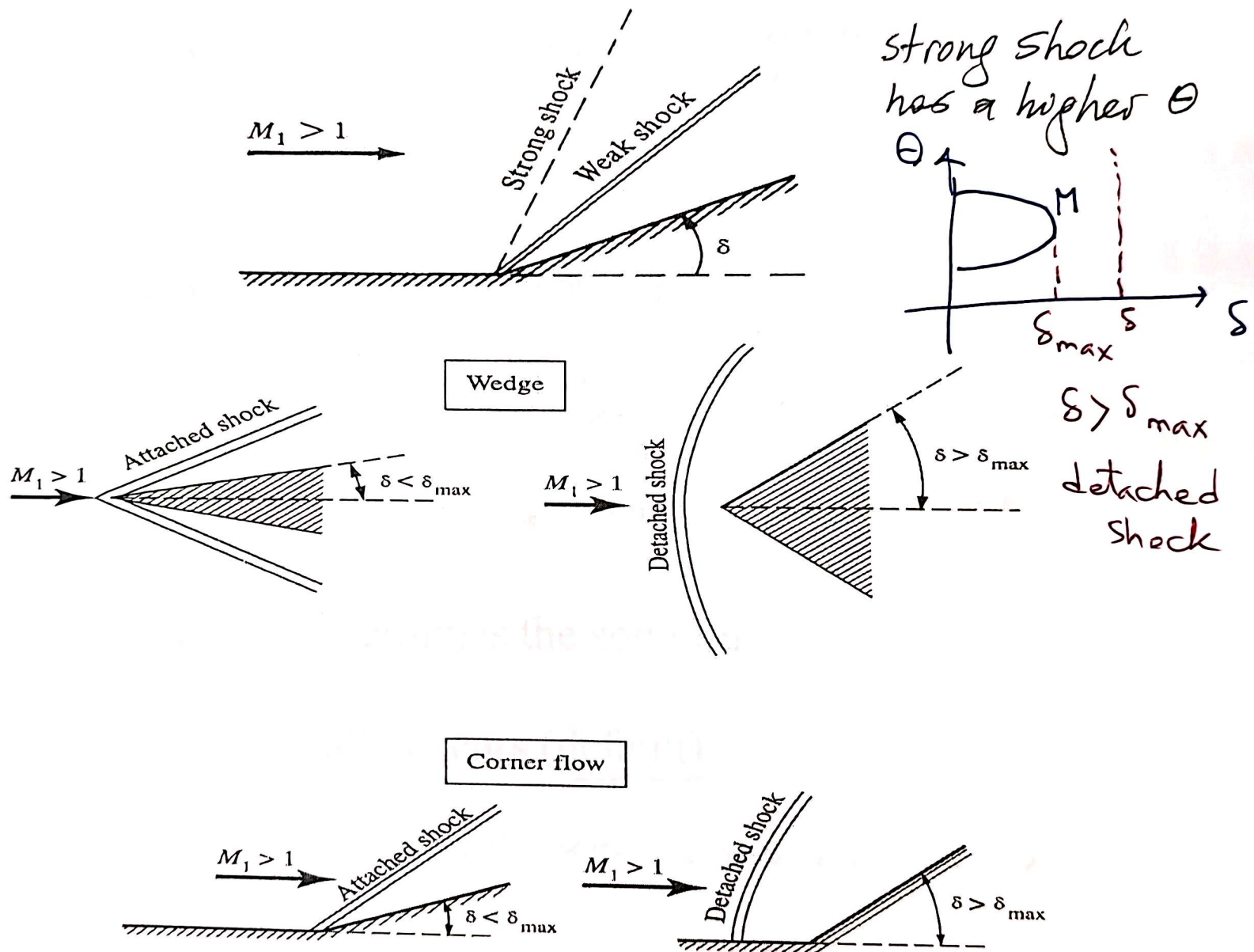


Comparison between isentropic and normal shock relations



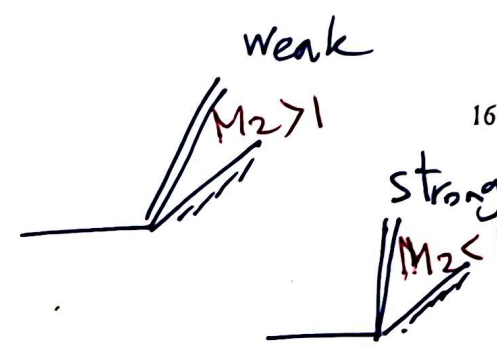
Weak shocks can be assumed isentropic.





for a strong shock $M_2 < 1$

for a weak shock $M_2 > 1$ (except near $\theta_{\max}$)



$\delta = 0 \Rightarrow \theta = \frac{\pi}{2}$ (strong) normal shock

$\theta = \mu$ (weak) Mach wave

- for a fixed δ as $M_1 \downarrow$ then $\theta \uparrow$ till the shock detaches.
(weak solution)
- as $M_{n1} \uparrow \Rightarrow$ stronger shock $\frac{P_2}{P_1} \uparrow$
- as $\delta \uparrow$ (with M_1 constant) $\Rightarrow$ stronger shock, $\theta \uparrow$ (weak)
- as $M_1 \uparrow$ (with δ constant) $\Rightarrow$ stronger shock, $\theta \downarrow$ (weak)

$$\begin{array}{l}
 M_1 = 3 \\
 P_1 = 1 \text{ atm} \\
 \delta = 2^\circ \\
 P_{t1} = 288 \text{ k}
 \end{array}$$

$$P_{t1}, P_2, T_2, M_2, P_{t2}, T_{t2}$$

$$\begin{array}{c}
 M=3 \\
 37.5^\circ = \theta \\
 \delta = 20
 \end{array}$$

$$P_2 = 1 \text{ atm}$$

$$C_p = \frac{P - P_\infty}{q_\infty}$$

$$q_\infty = \frac{1}{2} \rho_\infty U_\infty^2 = \frac{1}{2} \left(\rho_\infty M_\infty^2 \gamma R T_\infty \right)$$

$$\rho_\infty \gamma R T_\infty = P_\infty$$

$$\Rightarrow q_\infty = \frac{1}{2} P_\infty M_\infty^2$$